Project Title

Intelligent Optimal Route Planning Using Bit mask Dynamic Programming for the Travelling Salesman Problem

Project Report submitted for the partial fulfillment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Intelligent Optimal Route Planning Using Bit mask Dynamic Programming for the Travelling Salesman Problem project develops an efficient route-planning system for solving the Travelling Salesman Problem (TSP). The system determines the minimum-cost route that visits every city exactly once and returns to the starting city. Bit mask Dynamic Programming is used to efficiently represent visited cities and avoid repeated calculations, reducing the computational complexity compared with a brute-force approach. The system accepts a set of cities and their corresponding travel costs or distances, computes the optimal route, and displays both the selected path and its total travel cost. A route visualization component is also included to provide a clear graphical representation of the generated tour. The project demonstrates the practical application of bit masking, dynamic programming, state-space optimization, and combinatorial optimization in solving complex route-planning problems.

Signature of the Guide